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Article

SC-CO_2 and Brine Exposure Altering the Mineralogy, Microstructure, and Micro- and Macromechanical Properties of Shale

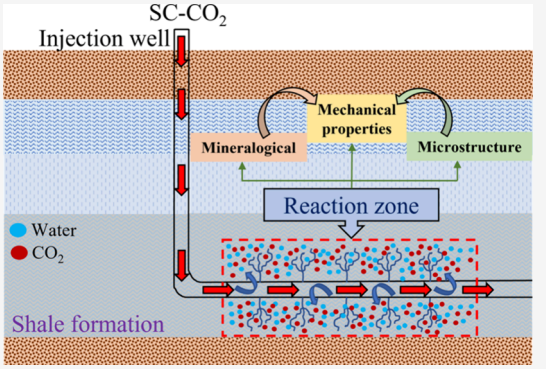
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ABSTRACT: Understanding the mechanism of the fluid−shale interaction, which is induced by SC-CO_2 injection, is critical to the safety of shale oil exploitation and the geological sequestration of CO_2. In this paper, the effects of different soaking fluids (SC-CO_2, brine, and SC-CO_2 + brine) and soaking time (7, 14, 21, and 28 days) on the physical properties of shale were investigated by performing static soaking experiments. A series of experimental methods (XRD, SEM, Nanoindentation, UCT, and Brazilian splitting) were used to analyze the variation of shale physical properties before and after soaking. The results showed that SC-CO_2 soaking had a limited effect on shale's physical properties. After soaking in brine, the degree of alteration in the physical properties of the shale increased. Furthermore, under the condition of SC-CO_2 + brine soaking, the mineral content and microstructure of shale changed most obviously. Following 28 days' soaking period, the microelastic modulus experienced an ultimate degradation extent of 44.99%, while the hardness experienced an ultimate degradation extent of 65.20%. The UCS, macroelastic modulus, and tensile strength were reduced by 47.23, 35.93, and 39.08%, respectively. Under the condition of SC-CO_2 + brine soaking, as the duration of soaking increased, the degree of variation of the shale mineral content intensified, the microstructure of shale altered more significantly, and the phenomenon of mineral dissolution and precipitation increased. Hence, as the soaking time increases, the deterioration of the micro-/macromechanical characteristics of the shale becomes more apparent, and the interaction between the soaking fluid and the shale becomes more obvious.



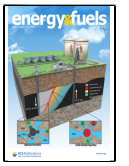
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1. INTRODUCTION

Relevant investigations have shown that supercritical carbon dioxide (SC-CO_2) can significantly enhance the recovery of shale reservoirs.[1−4] When the pressure and temperature of CO_2 reach a certain condition (generally $P \geq 7.38$ MPa, $T \geq 31.1$ °C), it will enter the supercritical phase. Thanks to its high density, minimal viscosity,[5] and excellent diffusivity,[6] supercritical carbon dioxide (SC-CO_2) easily mixes with crude oil and can effectively lower its viscosity.[7−9] This makes it easier for more shale oil to flow to the wellbore and efficiently enhances the recovery.[10,11] Meanwhile, CO_2 used can also be sequestered in shale reservoirs in the form of dissolution, trapping, and mineralization.[12−15]

The issue that the geochemical and physical interaction between SC-CO_2 and shale affects the shale oil yield and the stability of shale as a caprock cannot be ignored. Yin[16] et al. reported that SC-CO_2 has the effects of adsorption expansion, dissolution, and extraction, which can make the mineral content change alternately, the content of organic matter decrease, and the microstructure change. The alteration of the seepage channel caused by the change of microstructure is likely to have a beneficial impact on the production of oil.[17] The effect of SC-CO_2 on shale will lead to the deterioration of the mechanical properties of shale. Previous studies reported that the tensile strength,[18] triaxial compressive strength,[19] elastic modulus,[20] and fracture toughness[21] of shale deteriorated when interacting with SC-CO_2. The weakening of mechanical strength leads to the transformation of shale from brittle fracture to ductile failure.[22,23] In addition to investigating the influence of SC-CO_2 on the macromechanical changes of shale, scholars also did some research on the change law of micromechanical properties. Li[24] et al. studied the softening mechanism and action form of CO_2 on shale. At the same time, the micromechanical deterioration of shale after the interaction was studied. Cheng[25] et al. through nanoindentation experiments found that after SC-CO_2 treatment, shale high-hardness minerals and low-hardness minerals have deteriorated, but medium-hardness minerals have not softened. This provides a basis for the deterioration of

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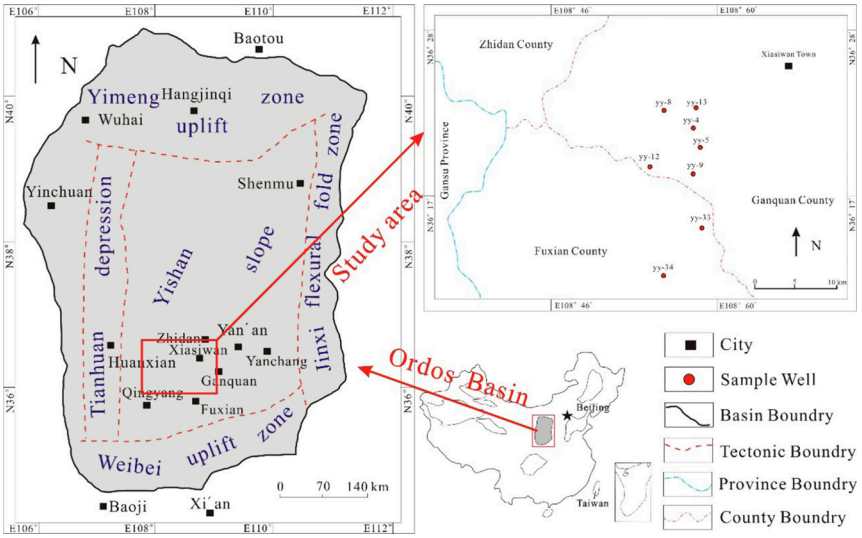


Figure 1. Map of the research area in the Ordos Basin, China. Reproduced from ref 39. Copyright 2016 American Chemical Society.

micromechanics. Moreover, the alteration of shale's physical properties is strongly correlated with the soaking time. Choi[26] et al. indicated that the softening of shale will occur, which is manifested in the decline of mechanical properties, and the decline range is negatively correlated with the time of exposure to the SC-CO_2 environment. Li[27] et al.'s investigation identified that the micromechanical characteristics of shale exhibit similar deterioration patterns. An investigation has been conducted on the mineral content and microstructure of shale samples that have been exposed to SC-CO_2 for long periods.[28] The findings indicate that after an extended period of static dissolution, the microstructure of shale has changed greatly, which is manifested in the development and growth of pores, the cracks became more complex, and the dissolution and precipitation of minerals intensified.

Indeed, shale oil reservoirs commonly contain mineralized water. Consequently, during CO_2 injection to extract shale oil, the fluid interacting with the shale is not only pure CO_2 but also pure brine and CO_2 + brine. In the presence of formation mineralized water, CO_2 can also react with formation water to generate an acidic solution.[6,29] Shale minerals have a larger probability of undergoing dissolution and precipitation in an environment with acidity, which can significantly alter the physical properties of shale reservoirs. Zhou[18] et al. compared the mechanical properties of shale specimens before and after soaking in different fluids (pure CO_2, pure brine, and combined soaking conditions). The findings indicated that the shale's mechanical properties deteriorated most significantly when exposed to a combination of CO_2 and brine soaking. Luo[30] et al. indicated that the main element content of shale changed alternately after interacting with saturated carbonate water and confirmed that the pore structure became complex by fractal dimension. The investigation of Zou[17] et al. confirmed that mineral dissolution occurred in shale exposed to CO_2-saturated saline environment, which significantly increased porosity and permeability and weakened the mechanical properties. Li[31] et al. concluded that the interaction of CO_2−brine−rock dissolves feldspar and carbonate minerals, resulting in a significant increase in the number of newly formed pores, increased pore radius, and connectivity. Furthermore, some scholars comprehensively discussed the influence of CO_2−brine−rock interaction on the alteration of shale deformation characteristics,[32] mechanical fracture characteristics,[33] macromechanical properties,[17,34] microstructure,[35−37] and micromechanical properties.[25,27,38]

However, most of the previous investigations only focused on the variations of pore structure, mineral composition, and macromechanical properties of shale before and after soaking. Currently, there is limited attention devoted to investigating the relationship among the variations in various properties of shale. Meanwhile, scholars have not given much attention to the changes in the physical properties of shale caused by exposure to SC-CO_2 + brine, which is the most probable scenario in actual production conditions. Therefore, to more comprehensively understand a series of geophysical−chemical reactions between injected SC-CO_2 and shale, the effects of different soaking fluids and soaking times on the physical properties of shale were investigated by performing static soaking experiments. A series of experimental methods (XRD, SEM, nanoindentation, UCT, and Brazilian splitting) were used to analyze the variation of shale physical properties before and after soaking. This investigation can reveal the mechanism of the alteration of the shale properties caused by SC-CO_2 and brine exposure.

2. MATERIALS AND METHODS

2.1. Specimen Description and Preparation. The shale specimens used in the experiment were all collected from the Triassic Yanchang Formation shale oil reservoir in the Ordos Basin, China, which is a typical continental shale oil reservoir (Figure 1). The shale specimens were crushed into about 200 mesh powders and mixed evenly for XRD experimental analysis. The polished samples were used for the nanoindentation experiments. It is noteworthy that the fragments used are all from the same shale, which effectively reduces the experimental error caused by the heterogeneity. Further, the shale beddings of all the fragments were perpendicular to the axial direction. The φ25 × 50 mm cylindrical specimen was used for the uniaxial compression test before and after soaking, and the φ50 × 25 mm cylindrical specimen was used for the Brazilian splitting test. The specimens were all processed from the same bulk shale. Moreover, the bedding of all cylindrical shale specimens is perpendicular to the axial direction, and there are no obvious cracks on the surfaces of the specimens.

2.2. Experimental Devices and Procedures. A high-temperature and high-pressure experimental apparatus was employed to conduct static soaking experiments on shale. The experimental setup, as depicted in Figure 2, primarily comprises a high-pressure drive pump, a

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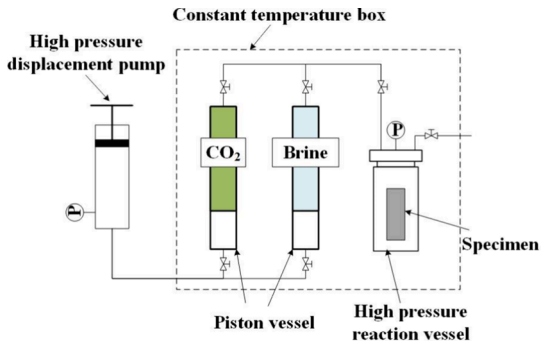


Figure 2. High-pressure and high-temperature static immersion test system.

constant-temperature box, two piston vessels, and a reaction vessel capable of withstanding pressures up to 50 MPa and temperatures as high as 100 °C.

The procedures of the static soaking experiment are depicted in Figure 3, and the soaking scheme is presented in Table 1. The soaking experiment was divided into seven groups, of which six groups were experimental groups with different soaking conditions and one group was treated as a control group without any treatment. Each group of experiments consisted of 100 g of crushed powder, one polished specimen, three $\varphi 25 \times 50$ mm, and three $\varphi 50 \times 25$ mm shale cylindrical specimens. The in situ shale oil reservoir conditions ($T = 67°C$, $P = 19$ MPa) were used as the soaking conditions. The brine used in the experiment is the formation water compounded in the laboratory, and its salinity is approximately 52.8 g/L. The specific ion composition is shown in Table 2. Before the static immersion experiment, the shale specimens were dehydrated in a 70 °C drying oven for 24 h. After being dried, the specimens were carefully placed inside the reaction vessel, and the powder samples were wrapped with a 1000 mesh gauze. The reaction vessel was evacuated to 0.09−0.1 MPa with a vacuum pump, and then the oven was opened and heated to 67 °C and maintained stable. The soaking fluid was pumped into the reaction vessel through a high-pressure drive pump and pressurized to 19 MPa for the experiments. After the specified soaking conditions were achieved, the valve was opened, the pressure slowly released, the soaked experimental samples were taken out, and the corresponding test was performed.

2.3. Analytical Techniques and Methods. *2.3.1. XRD and SEM Analyses.* The mineral content of shale powder specimens under various soaking environments was measured by a SmartLab X-ray powder diffractometer, and the scanning angle was 3°−65°. A Tescan Mira3 SEM instrument was utilized to observe the alterations of the microstructure of specimens.

2.3.2. Nanoindentation. The nanoindentation technique utilizes a diamond indenter with known mechanical properties to penetrate the surface of the specimen under a specific load (Figure 4a). Then, the contact area is indirectly defined by the depth of the indenter into the sample, and the relevant mechanical parameters are calculated. The microelastic modulus, hardness, creep properties, and fracture toughness of the specimen can be defined by the nanoindentation technique. Compared with traditional mechanical testing, nanoindentation technology has the advantages of simple, harmless, and rapid detection. The micromechanical properties (elastic modulus and hardness) of shale specimens before and after soaking were tested by a Hysitron Ti950 nanoindenter, with a maximum indentation depth of 1.5 mm and a displacement resolution of 0.02 mm. The maximum load used in the experiment was 5 mN. The micromechanical properties of the specimen can be determined from the load−displacement curve obtained by the experiment. A typical load−displacement curve (Figure 4b) is used as an example:[40]

$$H = \frac{P_{max}}{A_c} \tag{1}$$

$$A_c = 3\sqrt{3}h_c^2\tan^2\theta \tag{2}$$

where H represents the hardness, P_{max} is the maximum loading force, A_c is the contact area, h_c is the contact depth of the indenter penetrating the specimen surface, and $\theta = 65.3°$.[41]

$$E = (1-\nu^2)\left[\frac{1}{E_r} - \frac{1-\nu_i^2}{E_i}\right]^{-1} \tag{3}$$

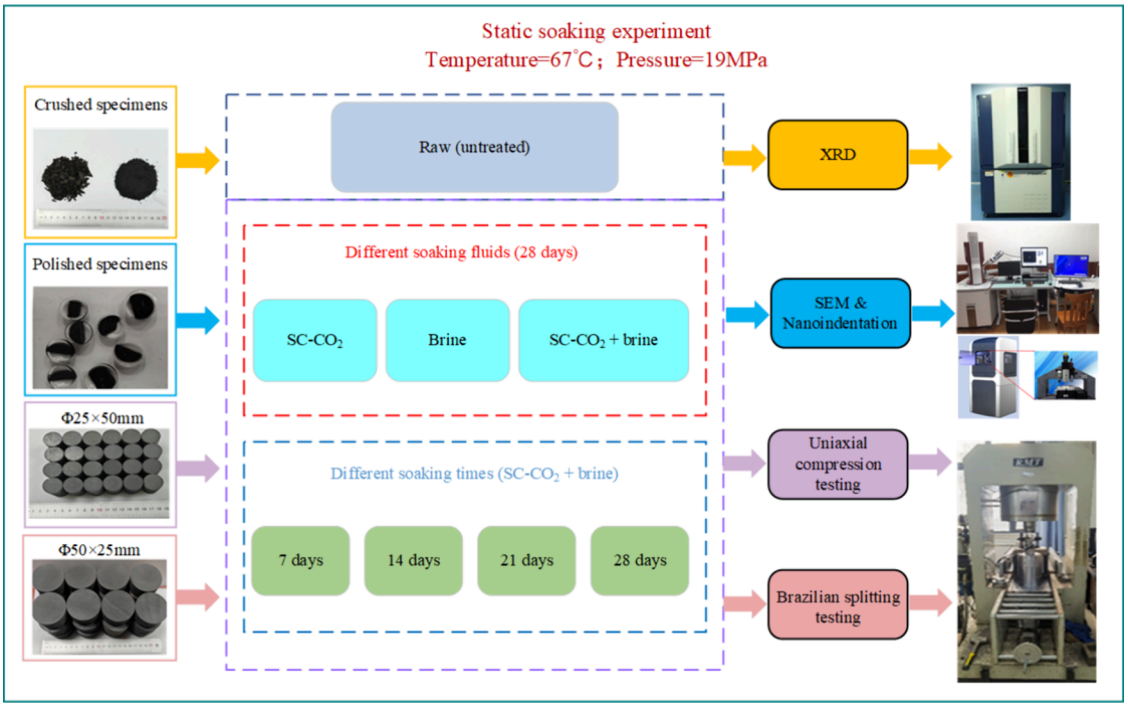


Figure 3. Experimental procedures and schemes.

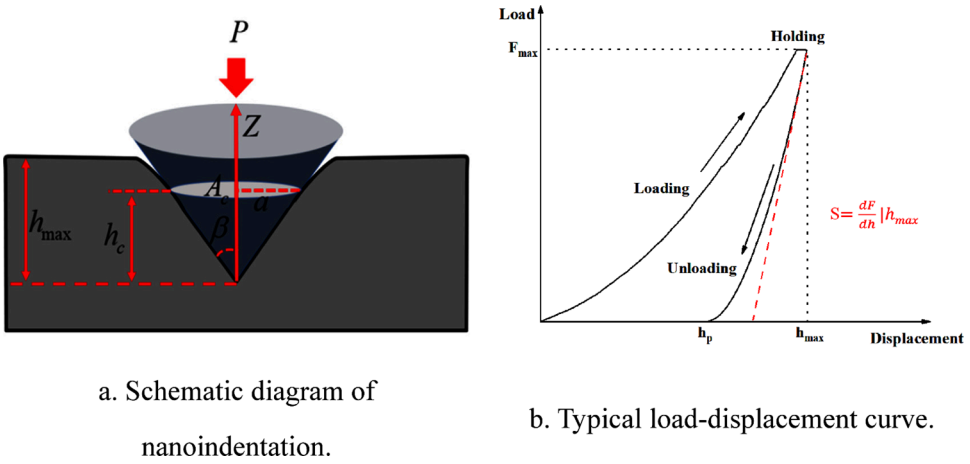
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Table 1. Soaking Experimental Scheme

sample ID	temperature (°C)	pressure (MPa)	reaction time (days)	soaking fluid	testing
#0	67	19	0	raw	
#1			28	SC-CO_2	XRD
#2			28	brine	SEM-EDS
#3			7	SC-CO_2 + brine	UCT
#4			14	SC-CO_2 + brine	Nanoindentation
#5			21	SC-CO_2 + brine	Brazilian test
#6			28	SC-CO_2 + brine	

Table 2. Ionic Composition of Brine

concentration (mg/L)						mineralization (mg/L)	type
Na^+, K^+	Ca^{2+}	Mg^{2+}	Cl^-	HCO_3^-	SO_4^{2-}	52843	$CaCl_2$
16350	4606	538	30601	256	492		



a. Schematic diagram of nanoindentation.

b. Typical load-displacement curve.

Figure 4. Schematic diagram of the nanoindentation theory.

Table 3. Mineral Compositions of Shale Under Different Soaking Fluids and Treatment Conditions

			mineral compositions (%)				
soaking fuid	reaction time (days)	soaking condition	quartz	feldspar[a]	carbonate[b]	clay[c]	pyrite
raw	0	19 MPa 67 °C	29.7	18.3	10.4	38.5	3.1
SC-CO_2	28		31.0	18.1	9.7	37.9	3.3
brine	28		32.9	18.6	8.8	36.4	3.3
SC-CO_2 + brine	7		34.8	17.0	7.9	36.2	4.1
SC-CO_2 + brine	14		35.2	17.3	6.5	37.3	3.7
SC-CO_2 + brine	21		36.9	15.6	6.8	36.1	4.6
SC-CO_2 + brine	28		40.1	15.8	4.7	34.4	5.0

[a]Feldspar minerals include k-feldspar and albite. [b]Carbonate minerals include calcite and dolomite. [c]Clay minerals include illite and chlorite.

$$E_r = \frac{\sqrt{\pi}}{2\beta\sqrt{A_c}}S \quad (4)$$

$$S = \frac{dP}{dh} \quad (5)$$

where E_i and ν_i are the elastic modulus and Poisson's ratio of the tip of the indenter, respectively. E_i = 1140 GPa, ν_i = 0.07, E and ν are the elastic modulus and Poisson's ratio of the test specimen, respectively, ν = 0.25,[42] E_r is the reduced modulus, S is the contact stiffness, and β is the geometric correction factor. For the Berkovich indenter, β = 1.034.[43]

2.3.3. Mechanical Testing. The uniaxial compression test and Brazilian splitting test were carried out using the RMT-301 rock mechanics experimental system. The uniaxial compression experiment was conducted with a loading rate of 0.001 mm/s, while the tensile experiment was conducted with a loading rate of 0.1 KN/s. The loading force continued to increase until the sample was destroyed. The data collection system was utilized to record the variations in stress and displacement within time.

3. RESULTS AND DISCUSSION

3.1. Alteration of the Mineral Content. The alterations of shale mineral composition before and after soaking were obtained by the XRD test, as shown in Table 3. The untreated shale samples are mainly composed of quartz, feldspar, carbonate, clay minerals, and pyrite, with contents of 29.7, 18.3, 10.4, 38.5, and 3.1%, respectively. The composition of feldspar is primarily made up of albite and K-feldspar, whereas carbonate is primarily composed of calcite and dolomite. Clay minerals, on the other hand, are mainly composed of illite and chlorite.

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The alternations in shale mineral composition induced by different soaking fluids (SC-CO_2, brine, and SC-CO_2 + brine) are depicted in Figure 5. It can be seen that after SC-CO_2, brine,

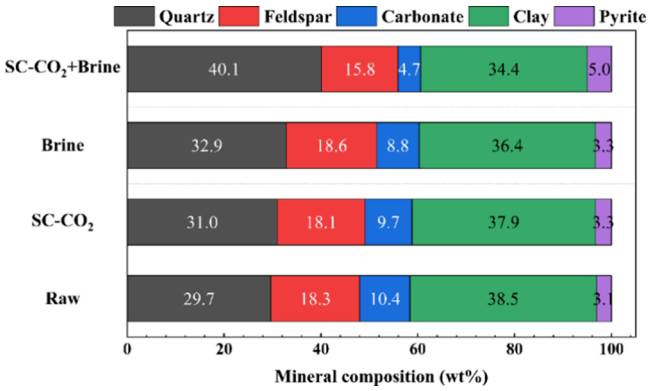


Figure 5. Mineral composition alternations of shale induced by different soaking fluids.

and SC-CO_2 + brine treatment, the content of quartz increased to 31.0, 32.9, and 40.1%, the content of carbonate decreased to 9.7, 8.8, and 4.7%, the clay mineral content decreased to 37.9, 36.4, and 34.4%, and the pyrite content increased to 3.3, 3.3, and 5.0%, respectively. Particularly, the content of feldspar minerals was reduced to 15.8% after the SC-CO_2 + brine soaking treatment. However, after the treatment of SC-CO_2 and brine, the content changes little. The alterations are due to differences in mineral solubility and the intensity of fluid−shale interactions. Therefore, under different soaking fluid conditions, the sequence of mineral dissolution intensity is SC-CO_2 + brine > brine > SC-CO_2. When the sample was exposed only to SC-CO_2, although some minerals reacted with pure CO_2, the dissolution effect was extremely limited. This is because it is difficult for SC-CO_2 to react with minerals under anhydrous conditions. However, SC-CO_2 can affect mineral composition by extracting interstitial water to undergo a series of chemical reactions with carbonate and clay minerals.[44] In the brine environment, the bicarbonate ions contained in the soaking solution provide a favorable acidic environment for the dissolution of minerals. Meanwhile, during the brine soaking procedure, the hydrolysis of shale occurred, resulting in the decomposition of carbonate and clay minerals, and their content decreased.[30,45] However, the hydrolysis effect was also relatively limited. It should be noted that when CO_2 is dissolved in brine, carbonic acid will be produced (eq 6). Accordingly, a series of chemical reactions occurred (eqs 7−14),[17,31,46,47] and the degree of mineral dissolution intensified.

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^- \quad (6)$$

$$CaCO_3(\text{Calcite}) + H^+ \rightleftharpoons Ca^{2+} + HCO_3^- \quad (7)$$

$$CaMg(CO_3)_2(\text{Dolomite}) + 2H^+ \rightleftharpoons Ca^{2+} + Mg^{2+} + 2HCO_3^- \quad (8)$$

$$\text{Illite} + 8H^+ \rightleftharpoons 0.25Mg^{2+} + 0.6K^+ + 2.3Al^{3+} + 3.5SiO_2(\text{Quartz}) + 5H_2O \quad (9)$$

$$\text{Chlorite} + 16H^+ \rightleftharpoons 2.3Al^{3+} + 3SiO_2(\text{Quartz}) + 5Fe^{2+} + 12H_2O \quad (10)$$

$$2KAlSi_3O_8(K-\text{feldspar}) + 2H^+ + 9H_2O \rightleftharpoons 2K^+ + 4H_4SiO_4 + Al_2Si_2O_5(OH)_4(\text{Kaolinite}) \quad (11)$$

$$NaAlSi_3O_8(\text{Albite}) + CO_2 + 5.5H_2O \rightleftharpoons Na^+ + HCO_3^- + 2H_4SiO_4 + 0.5Al_2Si_2O_5(OH)_4 (\text{Kaolinite}) \quad (12)$$

$$Fe^{2+} + 4SO_4^{2-} + 4H^+ \rightleftharpoons FeS_2(\text{Pyrite}) + 7O_2 + 2H_2O \quad (13)$$

$$SiO_2(\text{Quartz}) + H_2O \rightleftharpoons H_4SiO_4 \quad (14)$$

$$NaAlSi_3O_8(\text{Albite}) + H^+ \rightleftharpoons AlO(OH) + 3SiO_2(\text{Quartz}) + Na^+ \quad (15)$$

$$KAlSi_3O_8(K-\text{feldspar}) + H^+ \rightleftharpoons AlO(OH) + 3SiO_2(\text{Quartz}) + K^+ \quad (16)$$

Under various treatment conditions, the concentrations of quartz and pyrite increased, while the other minerals showed a decreasing trend, among which carbonate decreased the most. The dissolution sequence of minerals in an acidic solution is carbonate (eqs 7 and 8) > clay minerals (eqs 9 and 10) > feldspar (eqs 11 and 12) > quartz (eq 14), in which carbonate is the most easily dissolved and quartz is the most stable. Previous investigations have also confirmed the alternation patterns of these mineral compositions.[29,46,48] Under the acidic conditions of SC-CO_2 + brine, the dissolution of carbonates, clay minerals, and feldspar leads to a reduction in their relative proportion. Further, the phenomenon of an increase in quartz was related to the following: (i) Although quartz can be dissolved under high temperature and pressure, the degree was very weak (eq 14). As extensive other minerals react to dissolution, the total mineral mass decreases, resulting in a higher proportion of quartz. (ii) Illite and chlorite reacted with hydrogen ions under acidic conditions to produce substances containing quartz (eqs 9 and 10), which simultaneously increased the quartz content. (iii) Albite and K-feldspar can be converted to quartz in an acidic environment, which also increases the content of quartz in shale (eqs 15 and 16).[49,50] Similarly, the rise in pyrite proportion was correlated with the decline in the proportion of other minerals and may also be due to the increase of hydrogen ions to promote the precipitation of pyrite (eq 13).

As shown in Figure 6, after the SC-CO_2 + brine soaking treatment for 7, 14, 21, and 28 days, the clay mineral content was reduced by 2.3, 1.2, 2.4, and 4.1%, respectively. Conversely, the content of quartz increased by 5.1, 5.5, 7.2, and 10.4%, respectively. Carbonate tends to decompose in acidic environments, resulting in the greatest reduction in content. Further, the reduction was 2.5, 3.9, 3.5, and 5.7%, respectively. It can be found that as the soaking time increases, the mineral content alteration of the sample becomes more intense. The continuous change in mineral content over time is related to the following factors: (i) It takes time for the soaking fluid to enter the micropores and throats of the shale. (ii) The reaction will continue until the minerals are dissolved and saturated, which

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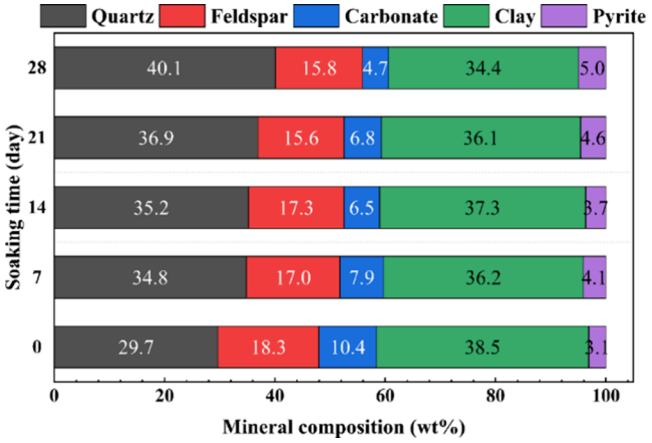


Figure 6. Mineral composition alternations of shale induced by different soaking times (under SC-CO_2 + brine exposure).

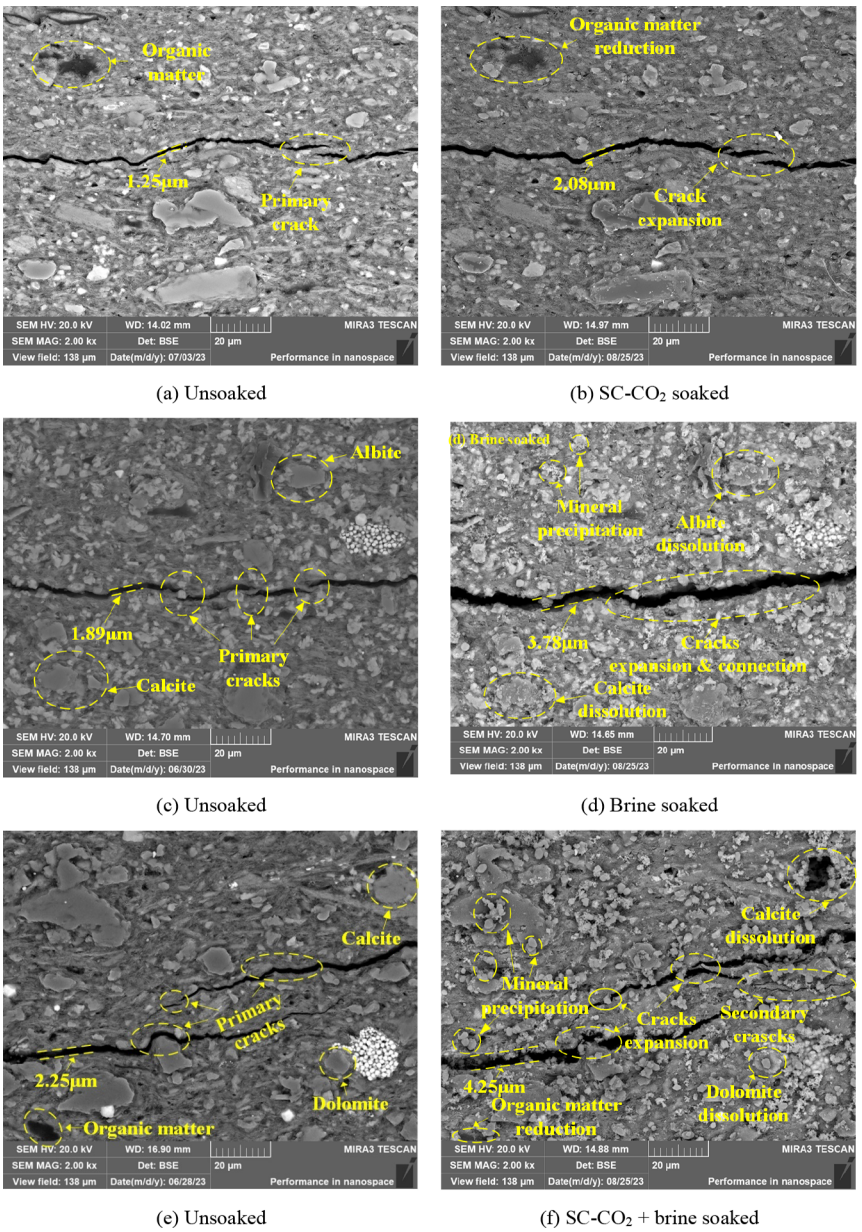
means that 28 days may not be enough to reach the saturation period. (iii) As the reaction proceeds, hidden minerals will appear and involve a dissolution process due to the erosion of the shale surface.

3.2. Alteration of Surface Morphology. The surface morphology of shale specimens treated with different soaking fluids was observed by SEM, as shown in Figure 7. After being soaked in SC-CO_2, the initial cracks on the shale surface were observed to undergo expansion, as depicted in Figure 7a,b. The main fracture width expanded from 1.25 to 2.08 μm, increasing by 66%. Simultaneously, significant alterations in pores and minerals were not observed, suggesting that the effect of CO_2 on shale was very weak, in terms of dissolving. The phenomenon can be explained by the fact that SC-CO_2 has an excellent affinity for shale and can be adsorbed on clay minerals to make them swell.[35] In the absence of water, it is difficult for CO_2 to react with inorganic minerals; therefore, there is no obvious dissolution of shale minerals. Meanwhile, SC-CO_2 can be used as a dehydrating agent to extract the bound water and interstitial water, causing mineral shrinkage.[51] Under the combined effect of the above factors, the shale has undergone uneven deformation. Interestingly, a reduction in organic matter was observed in Figure 7b. The emergence of this phenomenon is related to the fact that when the shale is soaked in SC-CO_2, SC-CO_2 acted as an organic solvent to dissolve and extract organic matter, resulting in a reduction in organic matter content.[52] In contrast to the SC-CO_2 soaking conditions, significant pore structure changes, mineral dissolution, and precipitation were observed on the surface of shale after brine treatment (Figure 7c,d). The initial main crack was expanded from 1.89 to 3.78 μm, and the previously unconnected cracks were also connected after soaking. Quartz did not dissolve after soaking in brine and still maintained its complete crystal structure. Conversely, both calcite and albite showed various degrees of dissolution reaction, and the dissolution of calcite was particularly significant. This aligns with the findings of XRD. Under the condition of brine soaking, hydrolysis is the main reason for mineral dissolution. In Figure 7e,f, it is observed that after the SC-CO_2 + brine soaking, the main fracture on the shale surface also expanded from 2.25 to 4.25 μm. Meanwhile, the propagation of secondary cracks was observed. Compared with Figure 7d, more mineral-precipitated crystals were observed in Figure 7f, indicating that more mineral dissolution occurred in shale. This can also be observed through the extent to which calcite and dolomite dissolve. After the soaking treatment, most of the calcite dissolved, and the content of dolomite reduced significantly, indicating that the mineral dissolution was extremely drastic. This is mainly due to the fact that the soaking fluid (SC-CO_2 + brine) not only has hydrolysis effect on shale minerals but also has carbonation effect, which can effectively dissolve minerals, and make the mineral dissolution and precipitation more intense.

Figure 8 illustrates the changes in the microstructure of the shale over time when exposed to SC-CO_2 + brine as a soaking environment. It is obvious that when the soaking time increases, the propagation behavior of the main crack gradually increases, from the initial single-crack propagation to multiple crack propagations and secondary crack derivation. This was related to the extent of the chemical reaction that occurred. Initially, the mineral dissolution was low, and the shale was dominated by SC-CO_2-induced adsorption expansion. With the gradual intensification of the reaction, the dissolution effect dominated, and the phenomena of crack extension and pore expansion occurred. In the first 2 weeks of soaking treatment, the alteration of shale surface morphology is mainly manifested as mineral dissolution, crack propagation, and extension. Significant mineral precipitation was observed on the 21st and 28th days of soaking. The reason for this phenomenon is that some ions dissolved in the soaking fluid will not precipitate before reaching saturation. As the soaking time increases, the ions tend to be saturated and gradually precipitate. Therefore, the precipitation phenomenon lags behind the dissolution phenomenon. In short, the more time the shale is soaked, the more the surface microstructure of the shale is changed.

3.3. Alteration of Micromechanical Properties. Shales are very heterogeneous rocks that exhibit significant variations in the mechanical characteristics of different minerals. As depicted in Figure 9, the loading displacement curve of shale nanoindentation is usually divided into three mechanical phases according to strength: soft phase, intermediate phase, and stiff phase.[53] The nanoindentation load−displacement curves of specimens after being soaked in various environments are depicted in Figure 9. After the soaking treatment, the softening amplitude of the soft and intermediate phases of shale was not drastical, while the softening amplitude of the stiff phase was large. The softening effect of different soaking fluids on shale is in the sequence SC-CO_2 + brine > brine > SC-CO_2. The load−displacement curve consists of three parts: loading stage, holding stage, and unloading stage, where the displacement of the holding stage reflects the creep properties of the shale surface.[54] In addition, discontinuous, step-like mutations were observed in the load−displacement curves, known as the "Pop-in" phenomenon. The occurrence of the "Pop-in" behavior is directly related to the microstructure of the shale. This phenomenon may occur when the indenter encounters microcracks, pores, and low-hardness minerals during loading, resulting in a sudden increase in displacement.[55−57] Therefore, the phenomenon of "Pop-in" can also be used as a quantitative indicator of the degree of shale softening. As shown in Figures 7 and 9, after static treatment with different soaking solutions, the surface micromorphology of shale has changed to varying degrees. It is characterized by fracture expansion and fracture connectivity, accompanied by a slight dissolution of minerals (albite and calcite). When the nanoindentation indenter is pressed into the surface of the soaked shale and contacts the cracks and dissolved minerals, the top limit of the diamond indenter is broken through instantaneously, resulting in a certain load and an increase in displacement. The increase in "Pop-in"

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(a) Unsoaked (b) SC-CO_2 soaked

(c) Unsoaked (d) Brine soaked

(e) Unsoaked (f) SC-CO_2 + brine soaked

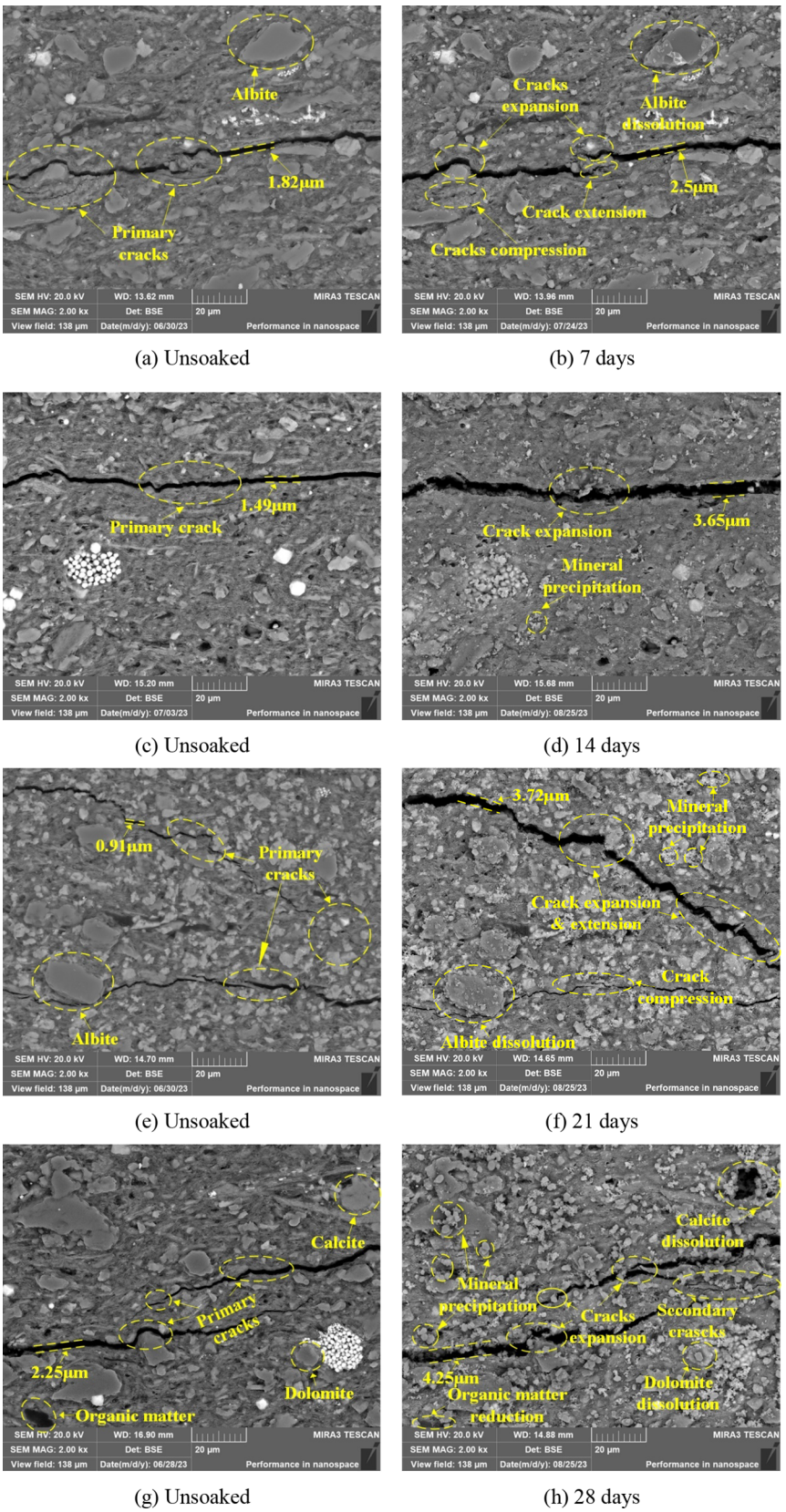
Figure 7. SEM images of the shale surface at different soaking fluids: (a) unsoaked sample; (b) SC-CO_2-soaked; (c) unsoaked sample; (d) brine-soaked; (e) unsoaked sample; (f) SC-CO_2 + brine-soaked.

events indicates that the shale surface contacted by the indenter is easier to break through and also indicates that the degree of shale softening and surface morphology change is greater. It can be found that after treatment with different soaking fluids, the degree of change in the micromorphology of shale is combined soaking > brine > pure SC-CO_2. Therefore, under the condition of combined soaking, the load−displacement curve is most prone to "Pop-in" events, followed by brine soaking. The holding stage of the load−displacement curve of shale increased significantly after SC-CO_2 + brine soaking, indicating that the creep properties of the shale changed significantly.

Figure 10 illustrates the variation of the nanoindentation load−displacement curve of the shale surface with soaking time when exposed to the conditions of SC-CO_2 + brine. As the duration of soaking increased, the intermediate and stiff phases of the shale were significantly softened, and the degree was positively correlated with the soaking time. Meanwhile, the holding stage of the load−displacement curve was getting longer with the increasing soaking time. This indicated that the creep properties of the shale were increasing, and the interaction between fluid and shale was also increasing. Another strong evidence that the soaking time had an effect on the softening behavior of shale is the increase of "Pop-in" phenomenon. According to the previous analysis, this is strongly associated with significant alterations in the shale pore structure. With the extension of soaking time, the microstructure of shale changes continuously, accompanied by severe expansion of main cracks and the formation of new microcracks. This means that the load required for the indenter to break through the shale surface is continuously reduced, and the probability of the "Pop-in" event is increased.

The statistical analysis of microelastic modulus and hardness data of shale treated with different fluids is shown in Figure 11. The high dispersion of the data point distribution reconfirmed

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(a) Unsoaked (b) 7 days

(c) Unsoaked (d) 14 days

(e) Unsoaked (f) 21 days

(g) Unsoaked (h) 28 days

Figure 8. SEM images of different soaking times under SC-CO_2 + brine treatment conditions: (a) unsoaked sample; (b) 7 days; (c) unsoaked sample; (d) 14 days; (e) unsoaked sample; (f) 21 days; (g) unsoaked sample; (h) 28 days.

the high heterogeneity properties of shale. The statistical law of micromechanical parameters was analyzed by normal distribution curve fitting. As illustrated in Figure 11, the microelastic modulus of shale, which was soaked by SC-CO_2, brine, and SC-CO_2 + brine, was reduced from 37.93 to 31.77, 26.61, and 20.86 GPa, respectively. The decreases were 16.24, 29.86, and 44.99%,

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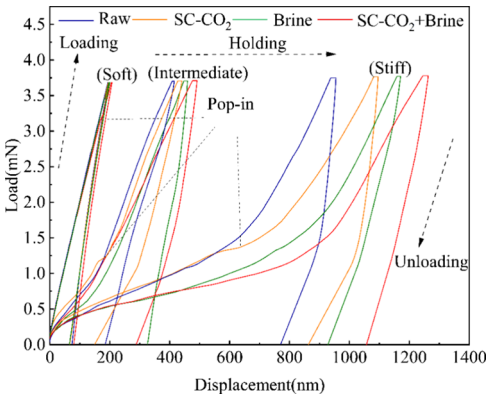


Figure 9. Load−displacement curves of nanoindentation in the shale treated with different soaking fluids.

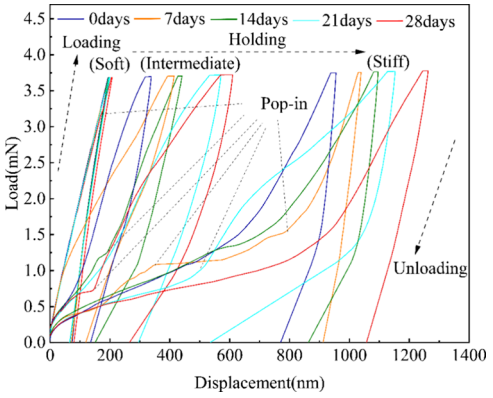


Figure 10. Load−displacement curves of nanoindentation in shale treated by different soaking times.

respectively. The variation of hardness was similar to that of the elastic modulus. After different soaking treatments of SC-CO_2, brine, and SC-CO_2 + brine, the hardness of shale was reduced from 3.42 to 2.70 2.55, and 1.19 GPa, respectively. The reduction was 21.05, 25.44, and 65.20%, respectively. The results indicated that the micromechanical properties of shale were altered most significantly under the condition of SC-CO_2 + brine. The phenomena and results observed by XRD and SEM can also be used to explain the change in micromechanical properties. Under the soaking conditions of SC-CO_2 + brine, the mineral content of shale altered most dramatically, and the dissolution effect is the most intense, indicating that the softening degree of shale was the most intense. Simultaneously, SEM observed that the cracks expanded the most, and the branch cracks increased the most, indicating that the microstructure of the shale altered greatly. The reduced surface energy of shale due to SC-CO_2 adsorption also deteriorates the surface performance. Under the combined effect of the above factors, the micromechanical properties of shale had been degraded.

Soaking time is also an essential factor, leading to shale softening. Figure 12 demonstrates that when exposed to the SC-CO_2 + brine condition, the variation of micromechanical parameters (elastic modulus and hardness) of shale was observed with soaking time. The elastic modulus of shale decreased from 37.93 to 34.62, 30.93, 27.53, and 20.86 GPa after 7, 14, 21, and 28 days of soaking treatment, respectively. The attenuation was 8.73%, 18.45, 27.43, and 44.99%, respectively. The hardness of the shale reduced from 3.42 to 3.03, 2.65, 1.48, and 1.19 GPa, respectively. The reduction was 11.40, 22.51, 56.73, and 65.20%, respectively. The micromechanical variation of shale was consistent with the results of XRD and SEM experiments. With the increase of soaking time, the influence of fluid on shale is more sufficient, the dissolution of minerals is more intense, and the degree of surface softening increases.

Particularly, compared to single SC-CO_2 treatment and pure brine treatment, the upper and lower limits of the mechanical characteristics' box plots of shale under the SC-CO_2 + brine combined soaking condition caused a significant shrinkage. Moreover, with the extension of soaking time, the range of the micromechanical box plot of the shale showed a similar shrinking trend. The occurrence of this phenomenon is most probably related to the drastic reduction of shale heterogeneity after soaking fluid treatment.

3.4. Alteration of Macromechanical Properties. The alterations in mechanical strength (including UCS, macroelastic modulus, and tensile strength) of specimens treated with various soaking fluids are depicted in Figure 13. After soaking in various fluids, the average UCS values of shale samples decreased by 10.99, 37.52, and 47.23%, respectively. The attenuation of elastic modulus was 5.41, 15.87, and 35.93%, respectively. Meanwhile, the tensile strength degeneration rates for SC-CO_2, brine, and SC-CO_2 + brine soaking specimens were 9.18, 20.14, and 39.08%, respectively. As illustrated in Figure 13a, the UCS of samples degenerated after exposure to different conditions, and the reduction sequence was SC-CO_2 + brine > brine > SC-CO_2. Figure 13b,c shows the alteration of tensile strength and elastic modulus similar to UCS. Accordingly, the experimental results indicated that the combined soaking environments of SC-CO_2 +

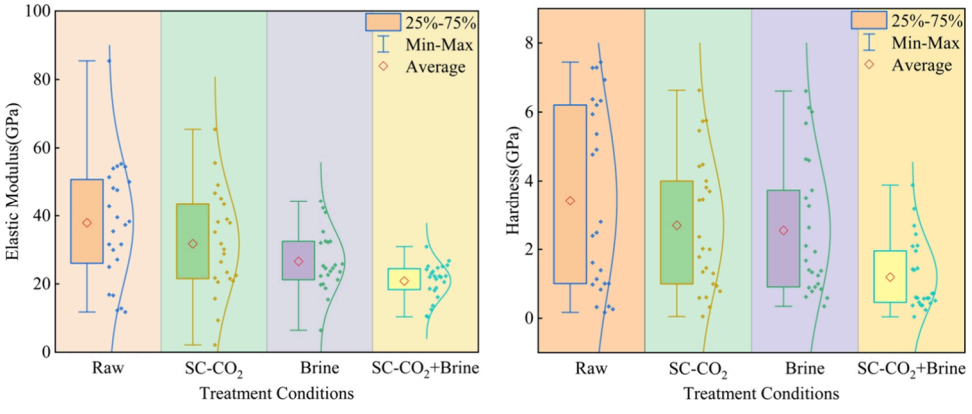


Figure 11. Elastic modulus and hardness of shale treated with different soaking fluids.

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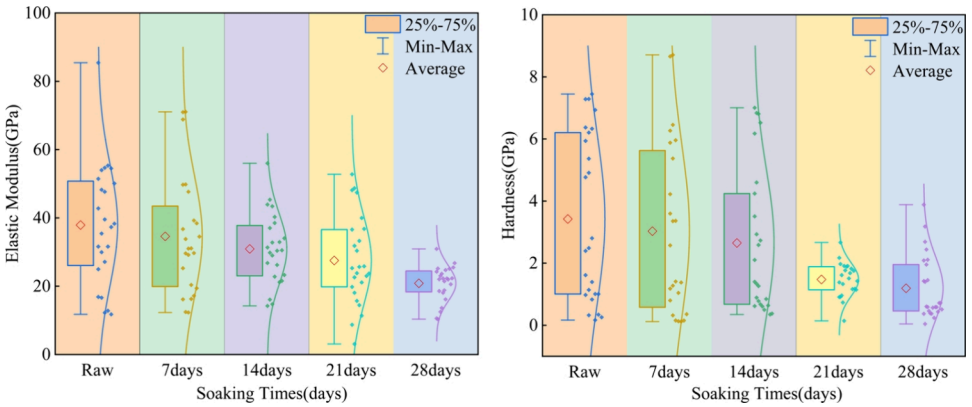


Figure 12. Elastic modulus and hardness of shale treated with different soaking times.

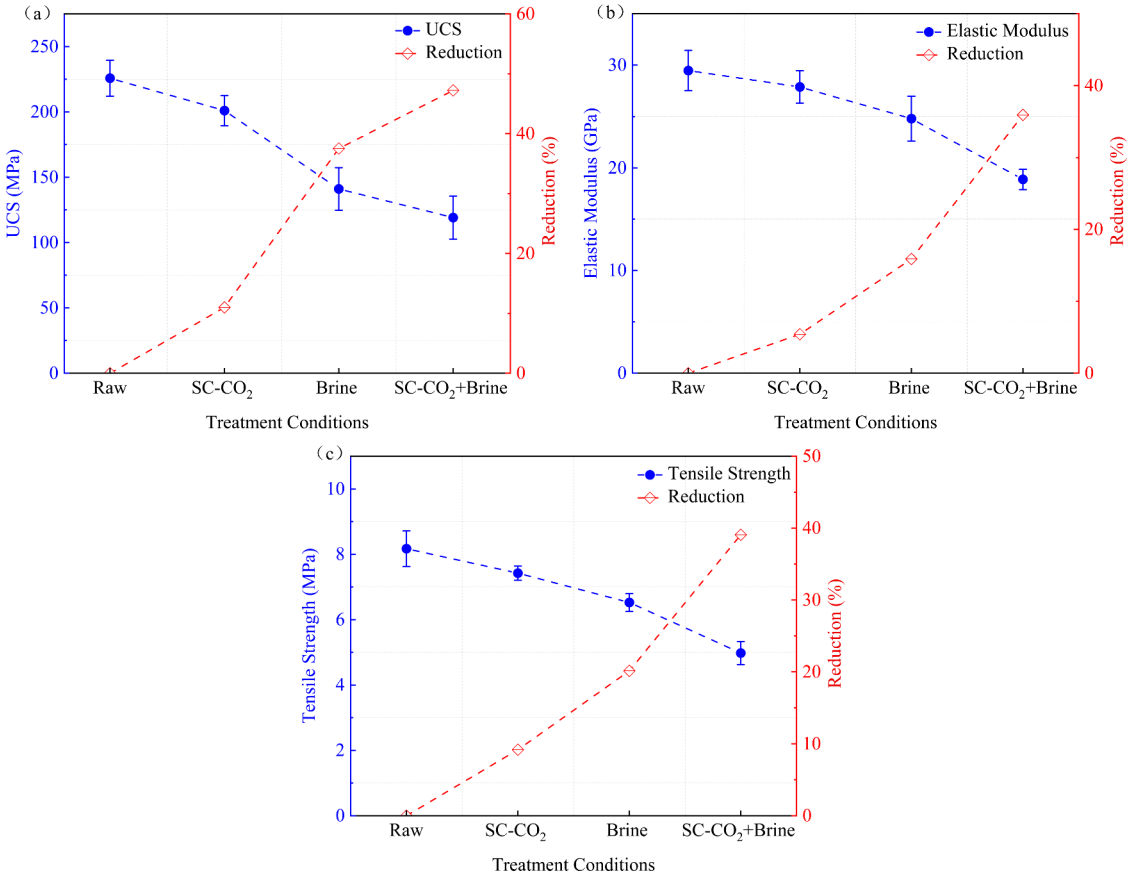


Figure 13. Variations in UCS, elastic modulus, and tensile strength of shale specimens after different soaking fluid treatments.

brine have a more significant softening effect on shale samples than pure SC-CO_2 and the corresponding pure brine treatment conditions.

The mechanical degradation of shale is regarded as the microstructural alterations induced by the exposure of the soaking fluid. However, different soaking fluids cause different degrees of damage to the microstructure of shale. As depicted in Figure 7b, the alterations caused by SC-CO_2 are limited. The mechanical performance of shale can be softened by pure SC-CO_2, which may be explained as follows: (i) SC-CO_2 will be adsorbed on shale, resulting in shale expansion and the appearance of secondary cracks.[34] (ii) The adsorption of CO_2 leads to a decrease in the surface energy of shale, resulting in a degradation of the stress required for crack damage and propagation.[58] (iii) SC-CO_2 dissolves and extracts organic matter to form pores, which promote the formation of shale microfractures and reduces the mechanical strength.[59] As depicted in Figure 7f, under the condition of SC-CO_2 + brine, the microstructure alternation caused by the interaction between shale and soaking fluid was the most significant, resulting in the most obvious attenuation of the shale mechanical strength. Moreover, the change of mineral content in the specimen is also an essential factor causing the alteration of mechanical properties. Compared with CO_2, brine has a stronger swelling effect on clay minerals and a hydrolysis effect on shale, which more significantly weaken the mechanical strength of shale.[18,60] Furthermore, the existence of water may cause the reduction of the cohesion and friction coefficient,

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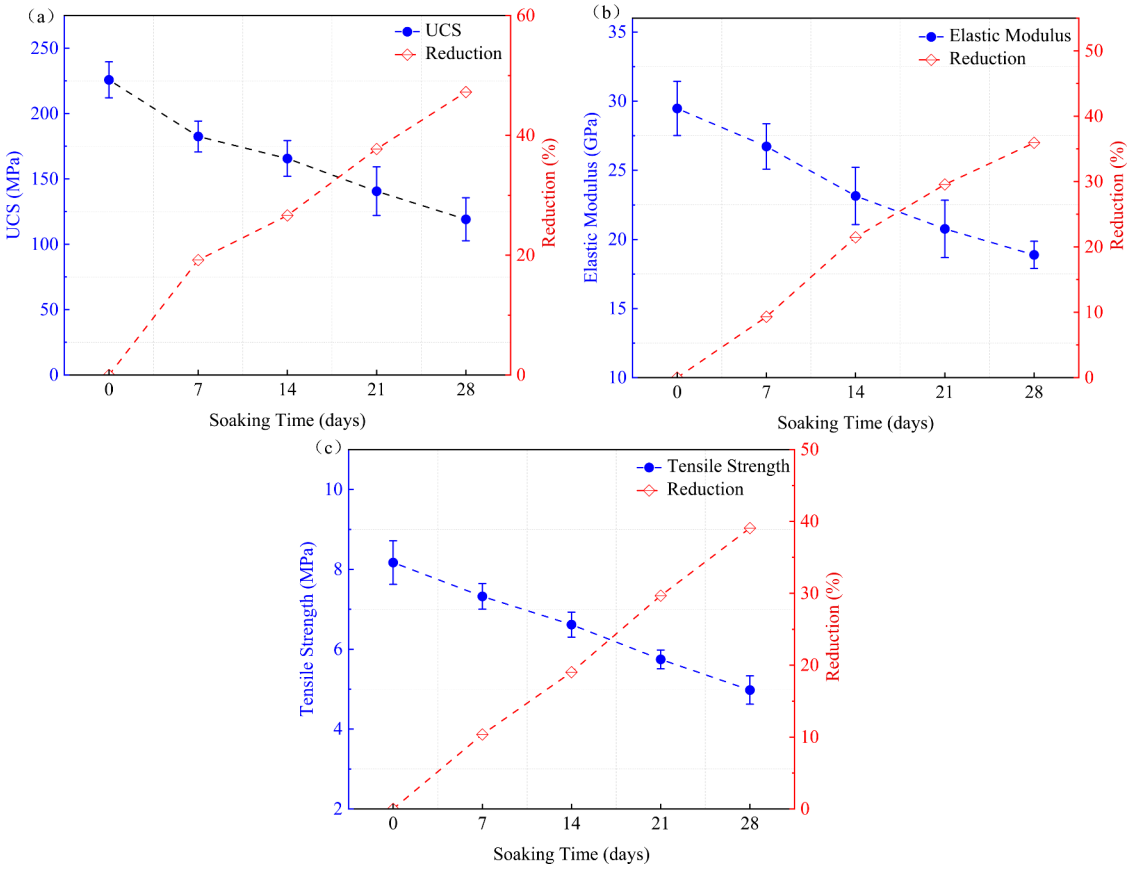


Figure 14. Variations in UCS, elastic modulus, and tensile strength of shale specimens after different soaking time exposures.

which would further weaken the strength of shale.[61,62] The soaking condition of SC-CO_2 + brine not only has the above weakening effect, but SC-CO_2 also produces carbonic acid by dissolving in water, which further strengthens the dissolution of minerals and enhances the erosion effect on the microstructure.

As shown in Figure 14, under the SC-CO_2 + brine soaking condition, the mechanical strength of shale is negatively correlated with the increase of the soaking time. After 7, 14, 21, and 28 days of soaking treatment, the UCS of shale decreased by 19.18, 26.64, 37.72, and 47.23%, respectively. After different soaking time treatments, the elastic modulus of shale decreased by 9.32, 21.48, 29.55, and 35.93%, respectively. Further, the tensile strength reduction rates were 10.36, 19.02, 29.68, and 39.08%, respectively. The alteration of shale microstructure is the primary factor influencing an alteration of mechanical performance. From the SEM results (Figure 8), it is observed that as the duration of soaking increased, the primary fractures continued to widen and the secondary fractures appeared. Under the combined erosion of SC-CO_2 + brine, shale minerals have undergone significant dissolution, resulting in the enlargement of shale pores and the decrease of adjacent fracture connectivity, which means a significant reduction of mechanical strength. Furthermore, with the increase of soaking time, more fluids can penetrate into the interior of the shale, resulting in the shale minerals continuously dissolving and the cracks continuously expanding and extending, so that the macromechanical strength continues to decrease.

Comparing the micro- and macromechanical experimental results of shale before and after soaking, it is found that the micro- and macromechanical degradation of shale caused by different soaking solutions and soaking times are similar. Specifically, after the SC-CO_2 soaking treatment, the shale mainly undergoes physical expansion, causing the microscopic increase in the fracture or changing the microscopic mechanical state between the mineral particles, thereby reducing the mechanical strength of the shale. It can be seen from the experimental results that after pure CO_2 soaking (Figures 11 and 13), the micro- and macromechanical properties of shale only decrease slightly, indicating that the physical mechanism has little effect on the change of micromechanical properties of shale. The macromechanical change is the result of the change in shale microstructure and mechanical state. Therefore, similar to the micromechanical properties, the macromechanical properties also show weak degradation. The environment of brine and combined soaking brings acidic chemical reaction conditions to shale, which makes shale further undergo a chemical mechanical degradation mechanism, so the micro- and macromechanical properties are greatly degraded. Especially under combined soaking, the reaction is more intense. The soaking time is positively correlated with the degree of the chemical reaction. The stronger the chemical action, the stronger the weakening effect of shale mechanical strength is, resulting in the weakening of micro- and macromechanical properties.

4. CONCLUSIONS

In this research, the physical properties of shale specimens under various soaking fluids (SC-CO_2, brine, and SC-CO_2 + brine) and soaking time (7, 14, 21, and 28 days) were investigated by using XRD, SEM, nanoindentation, uniaxial compression, and Brazilian splitting tests. The main results are as follows:

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1. When exposed to SC-CO_2, the alterations of shale mineral content and microstructure were extremely limited. After brine treatment, the shale surface cracks expanded, connectivity increased, and the variation of shale mineral content was intensified. Secondary cracks appeared after SC-CO_2 + brine soaking, and mineral dissolution and precipitation were intensified. Thus, the combined effect of SC-CO_2 + brine resulted in a more significant degree of alternation in the shale mineral content and surface morphology than both single SC-CO_2 and pure brine.
2. After soaking in SC-CO_2, brine, and SC-CO_2 + brine, the micromechanical strength of shale decreased significantly, the elastic modulus decreased by 16.24, 29.86, and 44.99%, and the hardness decreased by 21.05, 25.44, and 65.20%, respectively. The sequence of the weakening degree of shale micromechanical properties by different soaking fluids was: SC-CO_2 + brine > brine > SC-CO_2.
3. Compared with the raw shale specimens, after SC-CO_2, brine, and SC-CO_2 + brine treatment, the UCS of the shale decreased by 10.99, 37.52, and 47.23%, respectively. The average elastic modulus decreased by 5.41, 15.87, and 35.93%, respectively. Further, the average tensile strength decreased by 9.18, 20.14, and 39.08%, respectively. Therefore, SC-CO_2 + brine soaking conditions were more effective in reducing the mechanical strength than single SC-CO_2 or brine soaking conditions.
4. Under the condition of SC-CO_2 + brine soaking, increasing the duration of soaking time resulted in a greater degree of variation in the content of minerals in shale, the microstructure of shale altered more significantly, and the phenomenon of mineral dissolution and precipitation increased. Consequently, increasing the duration of soaking leads to a more significant deterioration of both the micromechanical and macromechanical characteristics of the shale, and the more intense the interaction between the soaking fluid and the shale becomes.

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Notes

The authors declare no competing financial interest.

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